

①

Admny Course
ML, NLP, PL

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Sec-22

Inferential Statistics

Lec-10 CHI Square
Test

Lec-11 CHI Square
Goodness of
FIT

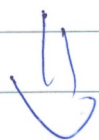
CHI Square Test :-

The CHI square test for the goodness of fit checks whether population proportions

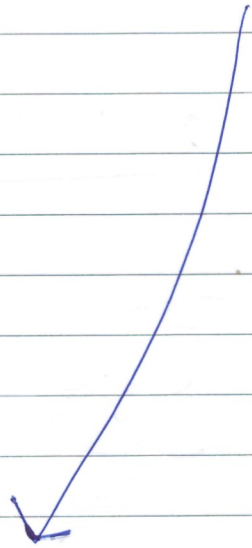
It is a non-parametric test that is performed on categorical data

④ There is a population of male who likes different color.

	Theory	sample
Yellow Bike	$1/3$	22
Red Bike	$1/3$	17
Orange Bike	$1/3$	59



Theory categorical distribution



observed categorical distribution

Goodness of Fit Test

- In a science class of 75 students, 11 are left handed.
- Does this class fit the theory that 12% of people are left handed?

		O	E
Obs	left handed	11	9
	right handed	64	66
		75	75

left handed = 11

right handed = 64

12% are left handed

$$\frac{12}{100} \times 75 = 9$$

When we talk about chi square test, we compare categorical values.

⇒ CHI square for goodness of fit.

In 2010, ~~ten~~ census of ~~the~~ city, the weight of the individuals in ~~the~~ small city were found to be following

$< 50 \text{ kg}$	$50 - 75$	> 75
207.	307.	507.

In 2020, the weight of $n = 500$ individuals were sampled.

< 50	$50 - 75$	> 75
140	160	200

(5)

Using $\alpha = 0.05$, would
you conclude the
population difference
of means has changed
in last 10 years?

Sol

2010
Expected
Data

<50 kg	50-75	>75
20%	30%	50%

2020
 $n = 500$
Observed

<50	50-75	>75
140	160	200

Expected

<50	50-75	>75
0.2×500 $= 100$	0.3×500 $= 150$	0.5×500 $= 250$

⇒ Null Hypothesis: H_0

→ The data meets the expectations.

① Alternate Hypothesis: H_1

• The data does not meet the expectations.

② $\alpha = 0.05$

$\therefore 1 - 0.05$
 $= 0.95$
 $= 95\%$

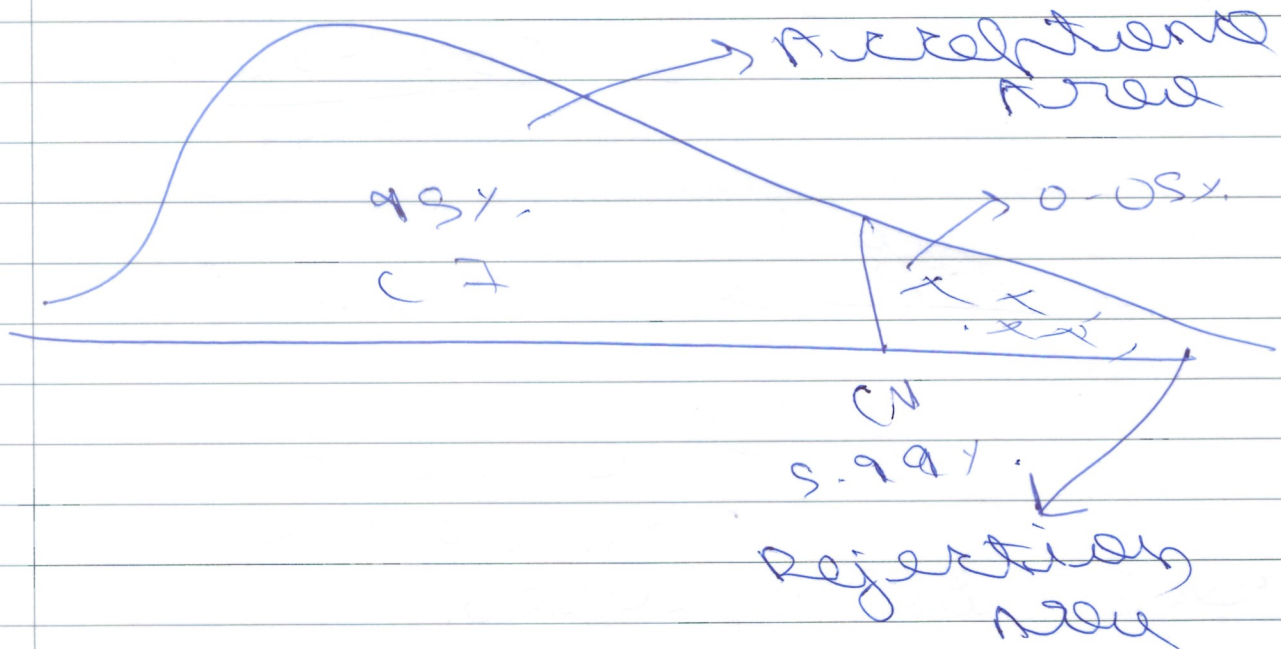
③ Degree of Freedom

$$df = K - 1 = 3 - 1 = 2$$

④ Decision Boundary:

7

⇒ Decision Boundary



If χ^2 is greater than 5.99, reject the null Hypothesis.

(5) Calculate the CHI Square Test Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

8

$$= \frac{(140-100)^2}{100} + \frac{(160-150)^2}{150} + \frac{(200-250)^2}{250}$$

$$= \frac{1600}{100} + \frac{100}{150} + \frac{2500}{250}$$

$$16 + 0.66 + 10 = 26.66 \text{ Ans}$$

same value used

2010	250	50-25	225
classified	140	160	200

Expected	250	50-25	225
	0.2×500 = 100	0.3×500 = 150	0.5×500 = 250

$$\chi^2 = 26.66$$

9

χ^2 is greater than
5.99. Reject H_0

also
we will reject the null hypothesis

26.66 > 5.99

→ reject the null hypothesis

Answer:-

The weights of 2020
population are different
than expected in
2010 population.